

BRAC UNIVERSITY
CSE331 : Automata and Computability
Assignment 1

1. Draw the state diagram of a DFA for the following RL:

$L_1(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 00\}$, where $\Sigma = \{0, 1\}$.

$L_2(M) \rightarrow \{w \in \Sigma^* \mid w \text{ doesn't contain } 11\}$, where $\Sigma = \{0, 1\}$.

A. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the length of } w \text{ is a multiple of 2 or 3}\}$, where $\Sigma = \{0, 1\}$. (use 6 states)

B. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the sum of the symbols of } w \text{ is a multiple of 3}\}$, where $\Sigma = \{0, 1, 2\}$.

C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the decimal equivalent of } w \text{ is a multiple of 5}\}$, where $\Sigma = \{0, 1\}$.

D. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 4 = 2\}$

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}$, where $\Sigma = \{0, 1\}$.

F. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{Every even position letter in } w \text{ is different from the first letter of } w\}$,
where $\Sigma = \{0, 1\}$.

G. $L(M) \rightarrow (L_1 \cap L_2)'$ (use 4 states)

2. Write a RE for the following RL:

$L_1(M) \rightarrow \{w \in \Sigma^* \mid \text{every third position in } w \text{ is } 1\}$, where $\Sigma = \{0, 1\}$.

$L_2(M) \rightarrow \{w \in \Sigma^* \mid \text{every } 1 \text{ in } w \text{ is followed by at least two } 0\}$, where $\Sigma = \{0, 1\}$.

A. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 3 \neq 1\}$, where $\Sigma = \{0, 1\}$.

B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$, where $\Sigma = \{0, 1\}$.

C. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of } 01 \text{ and } 10\}$, where $\Sigma = \{0, 1\}$.

D. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains at most two } 11\}$, where $\Sigma = \{0, 1\}$.

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ does not contain } 101\}$, where $\Sigma = \{0, 1\}$.

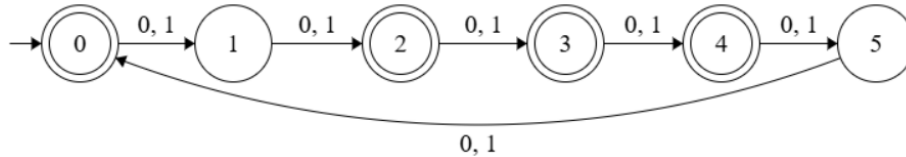
F. $L(M) \rightarrow L_1'$

G. $L(M) \rightarrow L_1 \cap L_2$

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Assignment 1

1. Draw the state diagram of a DFA for the following regular languages:

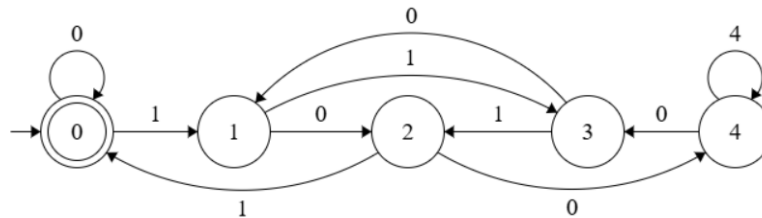
A. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the length of } w \text{ is a multiple of 2 or 3}\}$, where $\Sigma = \{0, 1\}$. (use 6 states)



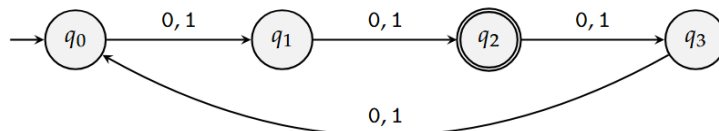
B. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the sum of the symbols of } w \text{ is a multiple of 3}\}$, where $\Sigma = \{0, 1, 2\}$.

See example 1.13 from the book.

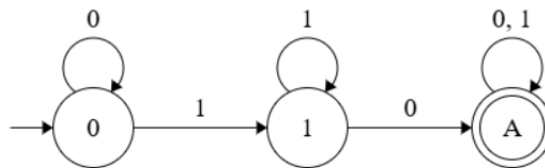
C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{the decimal equivalent of } w \text{ is a multiple of 5}\}$, where $\Sigma = \{0, 1\}$.



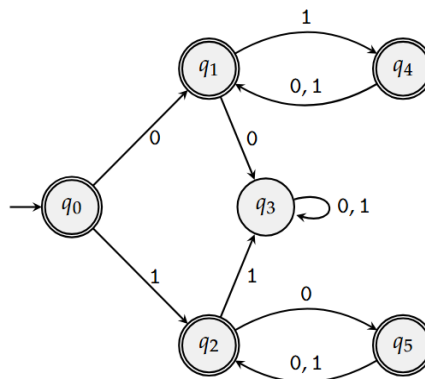
D. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 4 = 2\}$



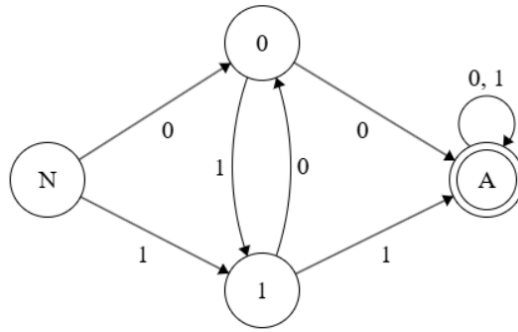
E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is any string not in } 0^*1^*\}$, where $\Sigma = \{0, 1\}$.



F. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{Every even position letter in } w \text{ is different from the first letter of } w\}$



G. $L(M) \rightarrow (L_1 \cap L_2)'$



2. Write the RE for the following regular languages:

A. $L(M) \rightarrow \{w \in \Sigma^* \mid |w| \% 3 \neq 1\}$

$(\Sigma\Sigma\Sigma)^*(\varepsilon \mid \Sigma\Sigma)$

B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ starts and ends with the same symbol}\}$, where $\Sigma = \{0, 1\}$.

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 0 \mid 1$

C. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of } 01 \text{ and } 10\}$, where $\Sigma = \{0, 1\}$.

$0\Sigma^*0 \mid 1\Sigma^*1 \mid 0 \mid 1 \mid \varepsilon$

D. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains at most two } 11\}$

$(0 \mid 10)^*(11 \mid \varepsilon)(0 \mid 10)^*(11 \mid \varepsilon)(0 \mid 10)^*(1 \mid \varepsilon)$

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ does not contain } 101\}$, where $\Sigma = \{0, 1\}$.

$(1|000^*)^*|\varepsilon|\Sigma|\Sigma\Sigma$

F. $L(M) \rightarrow L_1'$

$(\Sigma\Sigma\Sigma)^*\Sigma\Sigma 0\Sigma^*$

G. $L(M) \rightarrow L_1 \cap L_2$

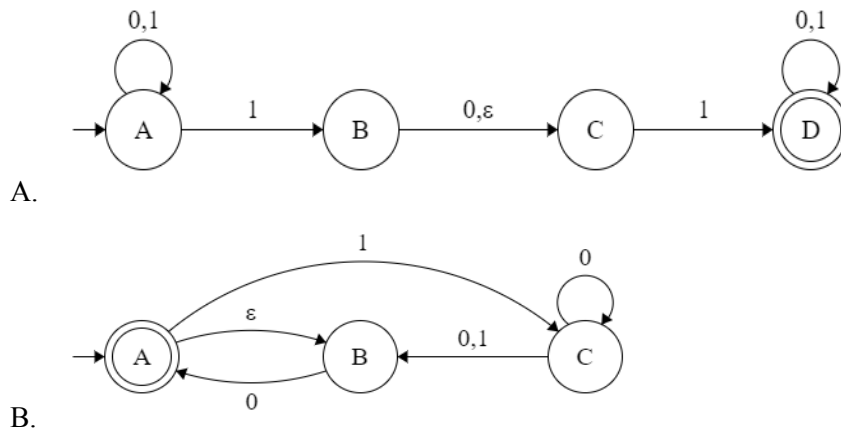
$(001)^*00 \mid \varepsilon \mid 0 \mid 00$

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Assignment 2

1. Draw the state diagram of an NFA for the following RL over $\{0, 1\}$:

- A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains } 1001 \text{ or } 11\}$. (use 5 states)
- B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains a } 1 \text{ in the third position from the end}\}$.
- C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{length of } w \text{ is a multiple of } 2 \text{ or } 3\}$.

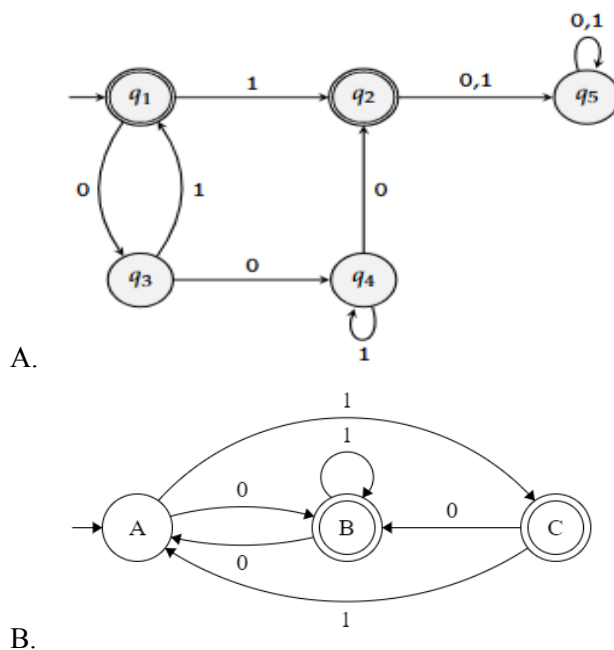
2. Convert the following NFA into DFA:



3. Convert the following RE into NFA:

- A. $10(01 \mid 0)^*$
- B. $(0 \mid 01^*0)^*01^*0$

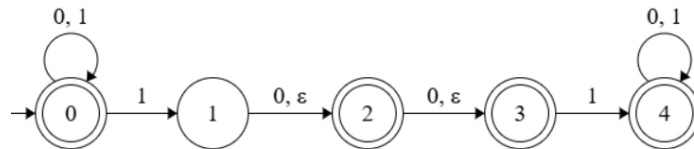
4. Convert the following DFA into RE:



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Assignment 2

1. Draw the state diagram of an NFA for the following regular languages:

A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains } 1001 \text{ or } 11\}$, where $\Sigma = \{0, 1\}$. (use 5 states)



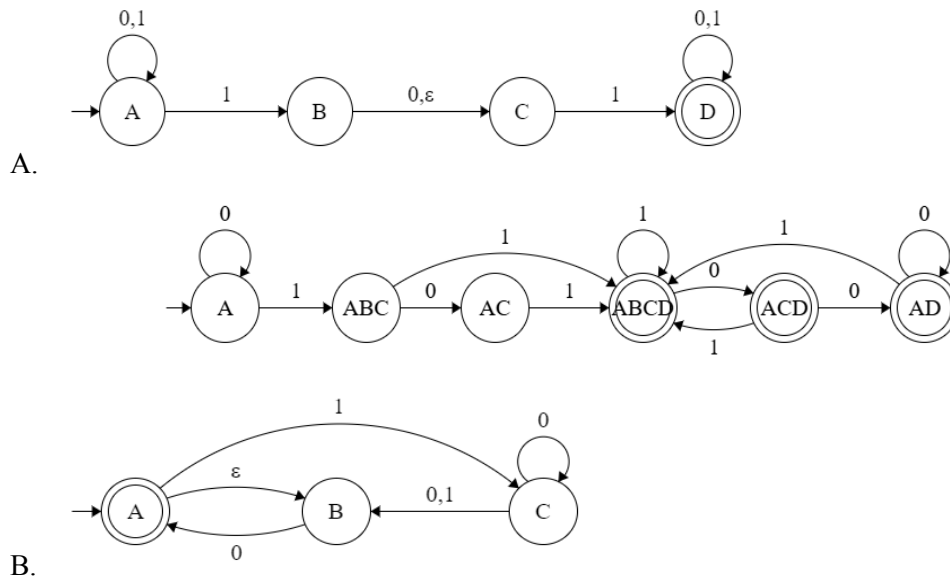
B. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains a } 1 \text{ in the third position from the end}\}$, where $\Sigma = \{0, 1\}$.

See example 1.30 from the book.

C. $L(M) \rightarrow \{w \in \Sigma^* \mid \text{length of } w \text{ is a multiple of } 2 \text{ or } 3\}$, where $\Sigma = \{0, 1\}$.

See example 1.33 from the book.

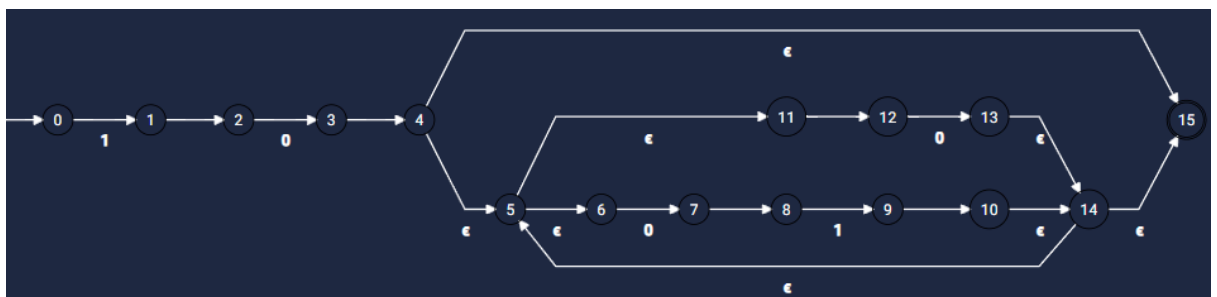
2. Convert the following NFA into DFA:



See example 1.41 from the book.

3. Convert the following RE into NFA:

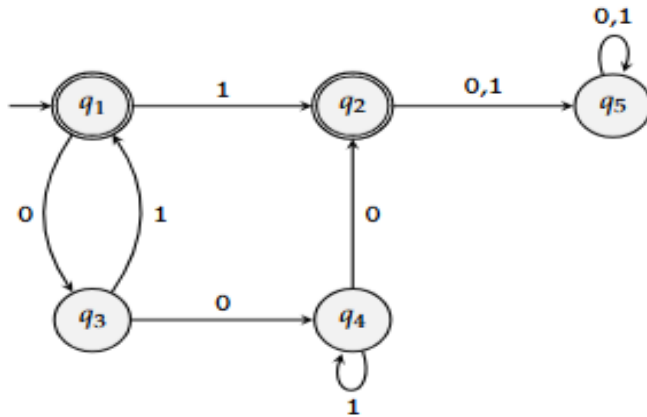
A. $10(01|0)^*$



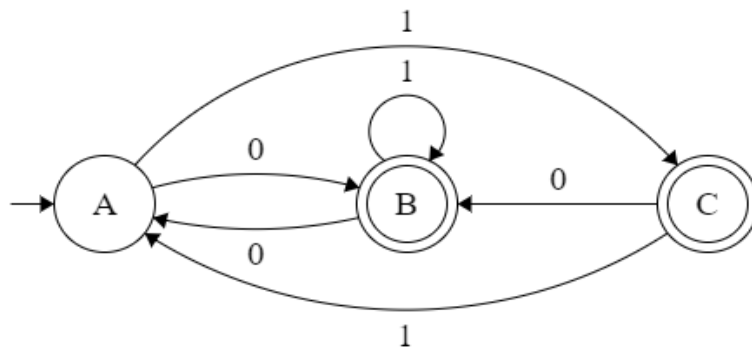
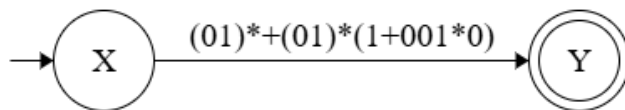
B. $(0 \mid 01^*0)^*01^*0$



4. Convert the following DFA into RE:



A.



B.

See example 1.68 from the book.

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Assignment 3

1. Use the pumping lemma to show that the following languages are not regular:

- A. $L(M) \rightarrow \{0^{2n}1^n \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.
- B. $L(M) \rightarrow \{0^n1^m0^{f(n,m)} \mid n, m \geq 0\}$, where $f(n, m) = n * m$ and $\Sigma = \{0, 1\}$.
- C. $L(M) \rightarrow \{0^n1^m \mid n > m\}$, where $\Sigma = \{0, 1\}$.
- D. $L(M) \rightarrow \{0^{n^2} \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.
- E. $L(M) \rightarrow \{w \text{ is not a palindrome}\}$, where $\Sigma = \{0, 1\}$.

2. Write a CFG for the following CFL:

- A. $L(M) \rightarrow \{0^n1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}$, where $\Sigma = \{0, 1\}$
- B. $L(M) \rightarrow \{0^n1^m2^m3^n \mid n, m > 0\}$, where $\Sigma = \{0, 1, 2, 3\}$
- C. $L(M) \rightarrow \{w = 0^i1^j2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}$, where $\Sigma = \{0, 1, 2\}$
- D. $L(M) \rightarrow \{w_1\#w_2 \mid \text{the number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}$, where $\Sigma = \{0, 1\}$
- E. $L(M) \rightarrow \{w\#x \mid w^R \text{ is a substring of } x\}$, where $\Sigma = \{0, 1\}$

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Assignment 3

1. Use the pumping lemma to show that the following languages are not regular:

A. $L(M) \rightarrow \{0^{2n}1^n \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.

See example 4 of the provided video resource on the Pumping Lemma for RL.

B. $L(M) \rightarrow \{0^n 1^m 0^{f(n,m)} \mid n, m \geq 0\}$, where $f(n, m) = n * m$ and $\Sigma = \{0, 1\}$.

See example 7 of the provided video resource on the Pumping Lemma for RL.

C. $L(M) \rightarrow \{0^n 1^m \mid n > m\}$, where $\Sigma = \{0, 1\}$.

See example 3 of the provided video resource on the Pumping Lemma for RL.

D. $L(M) \rightarrow \{0^{n^2} \mid n \geq 0\}$, where $\Sigma = \{0, 1\}$.

See example 10 of the provided video resource on the Pumping Lemma for RL.

E. $L(M) \rightarrow \{w \text{ is not a palindrome}\}$, where $\Sigma = \{0, 1\}$.

See example 18 of the provided video resource on the Pumping Lemma for RL.

2. Write a CFG for the following CFL:

A. $L(M) \rightarrow \{0^n 1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}$, where $\Sigma = \{0, 1\}$

$$S \rightarrow 000S11 \mid \epsilon$$

B. $L(M) \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}$, where $\Sigma = \{0, 1, 2, 3\}$

$$S \rightarrow 0S3 \mid A \mid \epsilon$$

$$A \rightarrow 1A2 \mid \epsilon$$

C. $L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}$, where $\Sigma = \{0, 1, 2\}$

$$S \rightarrow XABY \mid XA \mid BY$$

$$A \rightarrow 0A1 \mid \epsilon$$

$$B \rightarrow 1B2 \mid \epsilon$$

$$X \rightarrow 0X \mid 0$$

$$Y \rightarrow 2Y \mid 2$$

D. $L(M) \rightarrow \{w_1 \# w_2 \mid \text{the number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}$, where $\Sigma = \{0, 1\}$

$$S \rightarrow 0B1 \mid XS \mid SY \mid Z$$

$$X \rightarrow 1X \mid 01X \mid \epsilon$$

$$Y \rightarrow 0Y \mid 01Y \mid \epsilon$$

$$Z \rightarrow 0\# \mid \#1 \mid 0\#1 \mid \#$$

$$A \rightarrow 0B \mid 1A$$

$$B \rightarrow 0D \mid 1A$$

$$C \rightarrow D1 \mid C0$$

$$D \rightarrow E1 \mid 0C$$

$$E \rightarrow B \mid \#$$

E. $L(M) \rightarrow \{w\#x \mid w^R \text{ is a substring of } x\}, \text{ where } \Sigma = \{0, 1\}$

$$S \rightarrow AB$$

$$A \rightarrow 0A0 \mid 1A1 \mid \#B$$

$$B \rightarrow 0B \mid 1B \mid \varepsilon$$

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Assignment 4

1. Draw the state diagram of a PDA for the following CFL:

- A. $L(M) \rightarrow \{0^n 1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}$, where $\Sigma = \{0, 1\}$
- B. $L(M) \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}$, where $\Sigma = \{0, 1, 2, 3\}$
- C. $L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}$, where $\Sigma = \{0, 1, 2\}$
- D. $L(M) \rightarrow \{w_1 \# w_2 \mid \text{the number of } 00 \text{ in } w_1 \text{ is equal to the number of } 11 \text{ in } w_2\}$, where $\Sigma = \{0, 1\}$
- E. $L(M) \rightarrow \{w \# x \mid w^R \text{ is a substring of } x\}$, where $\Sigma = \{0, 1\}$

2. Draw the state diagram of a TM that decides the following languages:

- A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of } 0\text{s and } 1\text{s}\}$, where $\Sigma = \{0, 1\}$
- B. $L(M) \rightarrow \{0^n 1^n 2^n \mid n, m \geq 0\}$, where $\Sigma = \{0, 1, 2\}$
- C. $L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$, where $\Sigma = \{0, 1, 2\}$
- D. $L(M) \rightarrow \{0^{2^n} \mid n \geq 0\}$, where $\Sigma = \{0\}$
- E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$, where $\Sigma = \{0, 1\}$

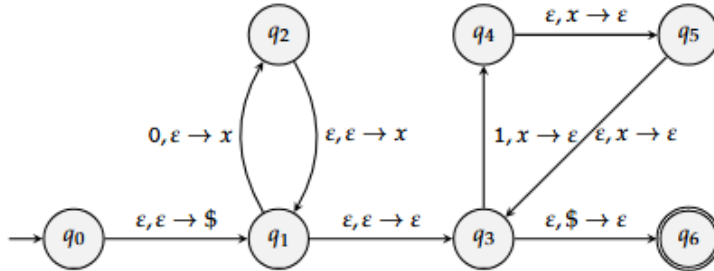
3. Prove that the following languages are decidable.

- A. $L = \{\langle M, w \rangle \mid \text{The Turing machine } M \text{ halts on input } w \text{ within } N \text{ steps}\}$
- B. $L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$
- C. Show that the collection of decidable languages is closed under Union.
- D. Show that the collection of Turing-recognizable languages is closed under Union.

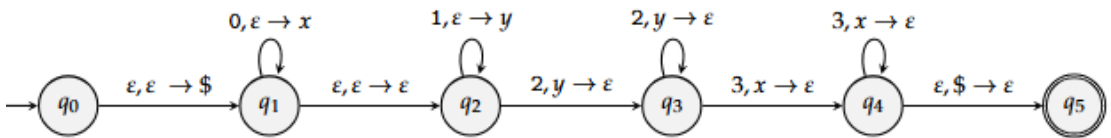
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Assignment 4

1. Draw the state diagram of a PDA for the following CFL:

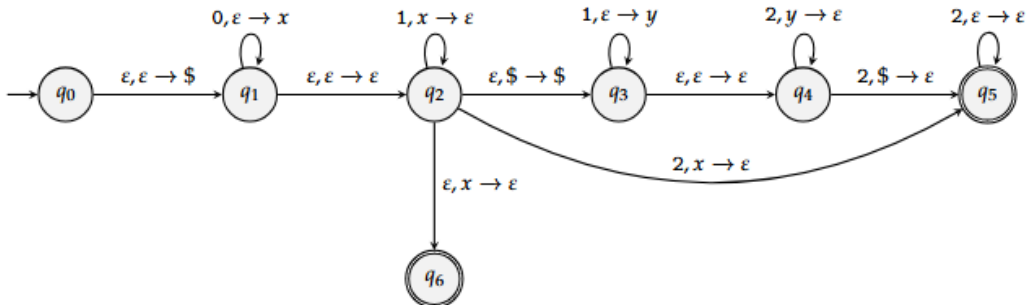
A. $L(M) \rightarrow \{0^n 1^m \mid n, m \geq 0 \text{ and } 2n = 3m\}, \text{ where } \Sigma = \{0, 1\}$



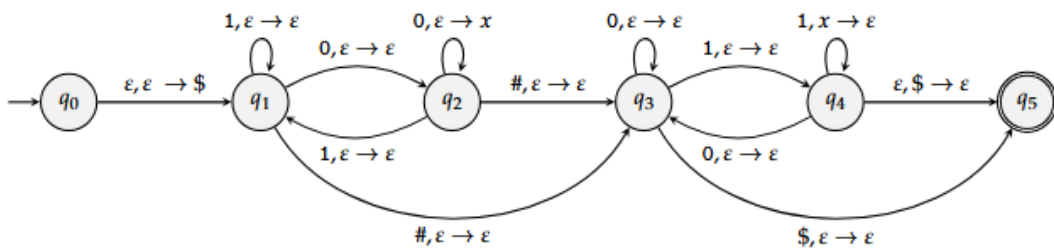
B. $L(M) \rightarrow \{0^n 1^m 2^m 3^n \mid n, m > 0\}, \text{ where } \Sigma = \{0, 1, 2, 3\}$



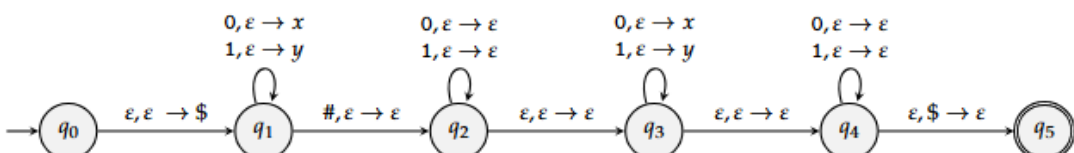
C. $L(M) \rightarrow \{w = 0^i 1^j 2^k \mid i, j, k \geq 0 \text{ and } j < i + k\}, \text{ where } \Sigma = \{0, 1, 2\}$



D. $L(M) \rightarrow \{w_1 \# w_2 \mid \text{the number of 00 in } w_1 \text{ is equal to the number of 11 in } w_2\}, \text{ where } \Sigma = \{0, 1\}$

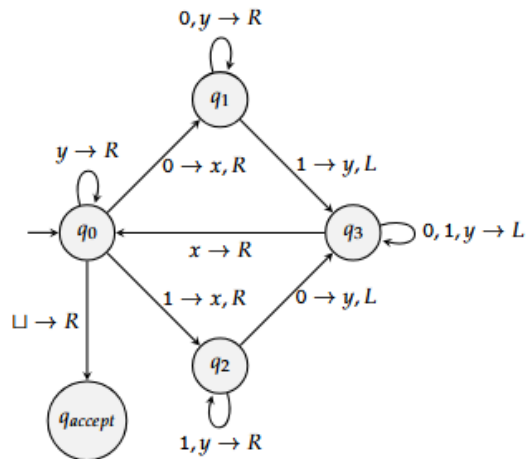


E. $L(M) \rightarrow \{w \# x \mid w^R \text{ is a substring of } x\}, \text{ where } \Sigma = \{0, 1\}$

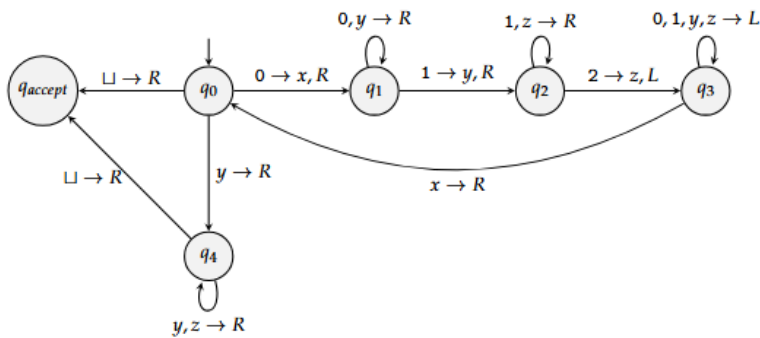


2. Draw the state diagram of a TM that decides the following languages:

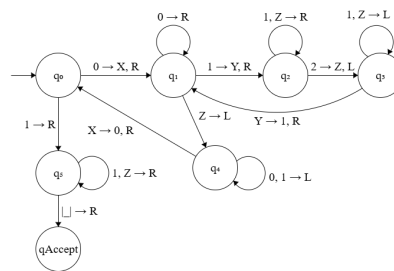
A. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ contains equal numbers of 0s and 1s}\}$, where $\Sigma = \{0, 1\}$



B. $L(M) \rightarrow \{0^n 1^m 2^n \mid n, m \geq 0\}$, where $\Sigma = \{0, 1, 2\}$



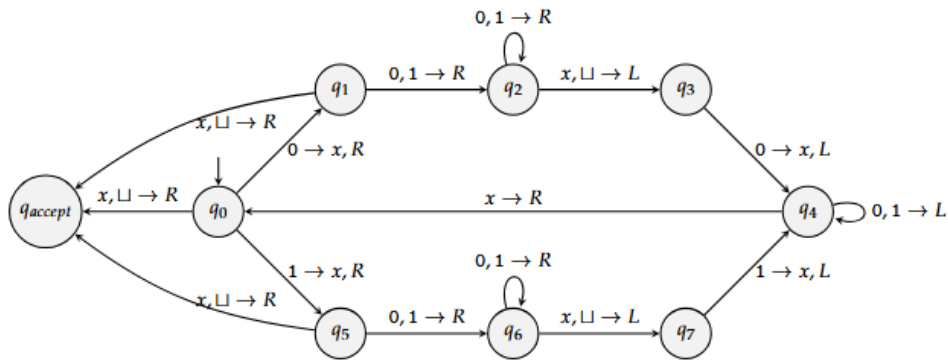
C. $L(M) \rightarrow \{0^i 1^j 2^k \mid i, j, k > 0 \text{ and } k = i * j\}$, where $\Sigma = \{0, 1, 2\}$



D. $L(M) \rightarrow \{0^{2^n} \mid n \geq 0\}$, where $\Sigma = \{0\}$

See example 3.7 from the book.

E. $L(M) \rightarrow \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$, where $\Sigma = \{0, 1\}$



3. Prove that the following languages are decidable.

A. $L = \{\langle M, w \rangle \mid \text{The Turing machine } M \text{ halts on input } w \text{ within } N \text{ steps}\}$

M_N : "On input $\langle M, w \rangle$:

1. simulate M on w for at most N steps.
2. If M halts within N steps, accept. Otherwise reject."

Since N is finite, M_N is a decider.

B. $L = \{\langle G \rangle \mid G \text{ is a connected undirected graph}\}$

See example 3.23 from the book.

4. Show that the following statements are true:

A. The collection of decidable languages is closed under Union.

See problem 3.15.a from the book.

B. The collection of decidable languages is closed under Complement.

For any decidable languages L , let M be the TM that decides L . We construct a TM M' that decides the complement of L .

M' ="On input w :

1. Simulate M on w .
2. If it accepts, accept. 2. Run M on w . If it accepts, accept. Otherwise, reject."

Since M is a decider, M' must be a decider.